

Toolpath-Guided Anisotropy for Elastic Wave Control: Theory and Algorithms for Orientation and Topology Optimization of Continuous-Fiber Composites

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Abstract

Continuous-fiber-reinforced polymer (CFRP) composites produced by additive manufacturing admit spatially varying, curvilinear fiber layouts that endow the material with a designable, locally anisotropic stiffness. This report gives a complete and self-contained mathematical description of a density- and orientation-based optimization framework for such composites, and of every numerical routine in its implementation. We document (i) the finite-element discretization and the *exact* finite Fourier representation of the rotated fiber stiffness that makes orientation-dependent assembly a scalar-weighted sum; (ii) the compactly supported radial-basis-function (CS-RBF) orientation map and its analytic spatial derivatives; (iii) the wave-projection “tool-path generation” operator and its discrete adjoint; (iv) the quasi-static compliance problem with local fiber-curvature (curl) constraints, reproducing Wong, Sanders & Rosen (*Compos. Struct.* 2026); (v) the time-harmonic elastodynamic solver with absorbing layers and the complex-symmetric adjoint used for wave-energy localization, a multi-frequency (broadband) max–min objective, and an elastic cloak; (vi) the Bloch band solver, Hellmann–Feynman eigenvalue sensitivities, and the gauge-invariant Fukui–Hatsugai–Suzuki Berry-curvature estimator; and (vii) the Method of Moving Asymptotes (MMA) and augmented-Lagrangian optimizers, with explicit update formulas. Every sensitivity is derived by the adjoint/Hellmann–Feynman method and verified against finite differences to $\sim 10^{-7}$. Numerical results demonstrate $> 20\times$ single-frequency energy focusing, uniform $\sim 14\times$ broadband focusing, and $\sim 15\times$ scattering reduction for a cloaked void, all driven purely by curved fiber toolpaths in an otherwise homogeneous plate. Finally, using the same orientation degree of freedom, we realize a *topological* valley-Hall edge waveguide: the rotation of a C_3 scatterer is a tunable Dirac mass whose sign sets the valley Chern number, and a domain wall between mirror-partner orientations guides a gap-traversing kink mode with a mid-gap/in-band transmission ratio of $\sim 10^3$. We further optimize the topological gap by MMA driven by Hellmann–Feynman sensitivities over the density and fiber-orientation fields, and give an honest accounting showing that in the anisotropic-fiber medium the orientation is a necessary co-ingredient of the optimized gap, not a passive accessory.

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1 Introduction

Topology optimization of fiber-reinforced composites has classically produced a density field together with a discrete fiber-orientation vector field, with the continuous manufacturing toolpath

generated only afterwards in post-processing [2, 3]. Because the as-manufactured fiber–matrix distribution differs from the homogenized material assumed during design, structural performance degrades in the transition to hardware. Recent *toolpath-integrated* formulations [1, 3] close this gap by generating an explicit fiber–matrix microstructure inside every optimization iteration and by constraining fiber curvature so that the design is manufacturable.

The same design freedom — a smooth, curvilinear, locally anisotropic stiffness tensor set by the fiber orientation — is a powerful lever for *elastic wave control*: whereas the quasi-static problem steers load paths, its dynamic counterpart steers energy paths. This document treats the density field $\mathbf{z}$ and the fiber-orientation field $\boldsymbol{\theta}$ as design variables and changes only the physics and objective between applications. The remainder is organized as a reference: Part I (§§2–6) develops the modeling primitives; Part II (§§7–12) the physics, objectives and their exact sensitivities; Part III (§§13–14) the optimization algorithms; Part IV (§15) the results; and §16 records verification data.

Throughout, N is the number of finite elements (and density variables), M the number of CS-RBF orientation support points, and n_{dof} the number of displacement degrees of freedom. Element quantities carry a subscript ℓ .

Part I: Modeling primitives

2 Finite-element discretization

The design domain is a rectangle $[0, L_x] \times [0, L_y]$ meshed by a regular grid of $n_{\text{elx}} \times n_{\text{ely}}$ bilinear quadrilateral elements of size $\Delta x \times \Delta y$. Each element ℓ has centroid $\mathbf{x}_\ell$ (a *design point*) and area $A_\ell = \Delta x \Delta y$. In the vector (in-plane) problem each node carries two displacement components; in the scalar (antiplane-shear) problem, one.

2.1 Shape functions and strain–displacement operator

On the reference square $(\xi, \eta) \in [-1, 1]^2$ the bilinear shape functions are

$$N_1 = \frac{1}{4}(1 - \xi)(1 - \eta), \quad N_2 = \frac{1}{4}(1 + \xi)(1 - \eta), \quad N_3 = \frac{1}{4}(1 + \xi)(1 + \eta), \quad N_4 = \frac{1}{4}(1 - \xi)(1 + \eta). \quad (1)$$

For an axis-aligned rectangle the Jacobian is constant, $\mathbf{J} = \text{diag}(\Delta x/2, \Delta y/2)$, $\det \mathbf{J} = \frac{1}{4}\Delta x \Delta y$, and physical gradients are $\partial_x N_a = \frac{2}{\Delta x} \partial_\xi N_a$, $\partial_y N_a = \frac{2}{\Delta y} \partial_\eta N_a$. The vector strain–displacement matrix $\mathbf{B} \in \mathbb{R}^{3 \times 8}$ (Voigt strains $[\varepsilon_{xx}, \varepsilon_{yy}, \gamma_{xy}]$) has the standard block form with $\mathbf{B}_{0,2a} = \partial_x N_a$, $\mathbf{B}_{1,2a+1} = \partial_y N_a$, $\mathbf{B}_{2,2a} = \partial_y N_a$, $\mathbf{B}_{2,2a+1} = \partial_x N_a$. The scalar gradient operator is $\mathbf{B}^s = \mathbf{J}^{-1}[\partial_\xi \mathbf{N}; \partial_\eta \mathbf{N}] \in \mathbb{R}^{2 \times 4}$.

2.2 Element matrices

With a 2×2 Gauss rule (points $\pm 1/\sqrt{3}$, unit weights) the element stiffness for a constitutive matrix $\mathbf{D}$, and the consistent mass, are

$$\mathbf{k}(\mathbf{D}) = \sum_g \mathbf{B}(\xi_g, \eta_g)^\top \mathbf{D} \mathbf{B}(\xi_g, \eta_g) \det \mathbf{J}, \quad \mathbf{m}(\rho) = \sum_g \rho \mathbf{N}_g^\top \mathbf{N}_g \det \mathbf{J}, \quad (2)$$

where $\mathbf{N}$ collects the shape functions into the 2×8 (vector) or 1×4 (scalar) interpolation matrix. Both maps are *linear* in their constitutive/density argument — the property exploited in §4. Global matrices are assembled by scatter–add over the element-DOF map, precomputed once as sparse row/column index arrays $(\mathbf{i}_K, \mathbf{j}_K)$.

3 Anisotropic constitutive model

3.1 In-plane (vector) fiber and matrix

In material axes the orthotropic fiber ply obeys plane-stress $\mathbf{Q}_f = \begin{bmatrix} Q_{11} & Q_{12} & 0 \\ Q_{12} & Q_{22} & 0 \\ 0 & 0 & Q_{66} \end{bmatrix}$ with

$$Q_{11} = \frac{E_{f1}}{1 - \nu_{12}\nu_{21}}, \quad Q_{12} = \frac{\nu_{12}E_{f2}}{1 - \nu_{12}\nu_{21}}, \quad Q_{22} = \frac{E_{f2}}{1 - \nu_{12}\nu_{21}}, \quad Q_{66} = G_{12}, \quad \nu_{21} = \nu_{12}\frac{E_{f2}}{E_{f1}}. \quad (3)$$

Rotating the ply to a physical angle θ uses the Voigt transformation

$$\mathbf{T}(\theta) = \begin{bmatrix} c^2 & s^2 & -2sc \\ s^2 & c^2 & 2sc \\ sc & -sc & c^2 - s^2 \end{bmatrix}, \quad c = \cos \theta, \quad s = \sin \theta, \quad (4)$$

giving the rotated fiber stiffness $\mathbf{D}_f(\theta) = \mathbf{T}(\theta) \mathbf{Q}_f \mathbf{T}(\theta)^\top$. The isotropic polymer matrix has plane-stress stiffness $\mathbf{D}_m = \frac{E_m}{1-\nu_m^2} \begin{bmatrix} 1 & \nu_m & 0 \\ \nu_m & 1 & 0 \\ 0 & 0 & (1-\nu_m)/2 \end{bmatrix}$.

3.2 Antiplane-shear (scalar) analogue

For out-of-plane displacement $u_z(x, y)$ the governing operator is $\nabla \cdot (\boldsymbol{\mu}(\theta) \nabla u_z) + \rho \omega^2 u_z = 0$ with a rank-2 shear tensor

$$\boldsymbol{\mu}(\theta) = \mathbf{R}(\theta) \text{diag}(\mu_L, \mu_T) \mathbf{R}(\theta)^\top = a \mathbf{I} + b \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}, \quad a = \frac{\mu_L + \mu_T}{2}, \quad b = \frac{\mu_L - \mu_T}{2}, \quad (5)$$

where $\mathbf{R}(\theta)$ is the 2×2 rotation. The scalar element stiffness is $\mathbf{k}^s(\boldsymbol{\mu}) = \sum_g (\mathbf{B}^s)^\top \boldsymbol{\mu} \mathbf{B}^s \det \mathbf{J}$.

4 Exact finite Fourier representation of orientation

Because $\mathbf{T}(\theta)$ in (4) has entries that are degree-2 monomials in (c, s) , each entry of $\mathbf{D}_f(\theta) = \mathbf{T} \mathbf{Q}_f \mathbf{T}^\top$ is a trigonometric polynomial of degree at most four, i.e. an exact linear combination of the five basis functions

$$\mathbf{c}(\theta) = [1, \cos 2\theta, \sin 2\theta, \cos 4\theta, \sin 4\theta]. \quad (6)$$

Since the element map (2) is linear in $\mathbf{D}$, the element stiffness inherits the *exact* expansion

$$\mathbf{k}_f(\theta) = \sum_{s=0}^4 \mathbf{k}_{f,s} c_s(\theta), \quad \mathbf{k}_{f,s} = \mathbf{k}(\mathbf{C}_s), \quad (7)$$

where $\mathbf{C}_s$ are the 3×3 coefficient matrices of $\mathbf{D}_f$ in the basis (6). In the implementation the $\mathbf{C}_s$ are recovered by an (overdetermined, exactly solvable) least-squares fit of $\mathbf{D}_f(\theta)$ sampled densely over $\theta \in [-\pi, \pi)$; the residual is $\mathcal{O}(10^{-16})$, confirming exactness. The four coefficient element matrices $\mathbf{k}_{f,s}$ and the matrix-phase element stiffness $\mathbf{k}_m = \mathbf{k}(\mathbf{D}_m)$ are precomputed *once*. During optimization, assembly of $\mathbf{k}_f(\theta_\ell)$ and its derivative reduce to

$$\mathbf{k}_f(\theta) = \sum_s \mathbf{k}_{f,s} c_s(\theta), \quad \frac{d\mathbf{k}_f}{d\theta} = \sum_s \mathbf{k}_{f,s} c'_s(\theta), \quad (8)$$

with $\mathbf{c}'(\theta) = [0, -2\sin 2\theta, 2\cos 2\theta, -4\sin 4\theta, 4\cos 4\theta]$. The scalar case is the exact three-term analogue with basis $\{1, \cos 2\theta, \sin 2\theta\}$ and closed-form coefficients $\mathbf{C}_0 = a\mathbf{I}$, $\mathbf{C}_1 = b \text{diag}(1, -1)$, $\mathbf{C}_2 = b \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ from (5). This precomputation is the key to making orientation-dependent assembly (and its derivative) $\mathcal{O}(N)$ per iteration.

5 Density and orientation parametrization

5.1 Density: filter and SIMP

The density variables $\mathbf{z} \in [0, 1]^N$ are regularized by a linear (cone) filter $\mathbf{y} = \mathbf{P}\mathbf{z}$, where

$$P_{ij} = \frac{w_{ij}A_j}{\sum_k w_{ik}A_k}, \quad w_{ij} = \max(0, R - \|\mathbf{x}_i - \mathbf{x}_j\|), \quad (9)$$

with filter radius R . For periodic (tileable) unit cells the distance in (9) is the minimum-image distance on the torus, so that the physical density is seamlessly periodic. Intermediate densities are penalized by the SIMP interpolation $E_\ell = \varepsilon + (1 - \varepsilon)y_\ell^p$, with penalty $p > 1$ and Ersatz floor ε .

5.2 Orientation: CS-RBF map and analytic spatial derivatives

Orientation design variables $\hat{\boldsymbol{\theta}} \in [-\frac{\pi}{2}, \frac{\pi}{2}]^M$ live at M support points $\mathbf{x}_q$ and are interpolated to element angles $\boldsymbol{\theta} = \hat{\boldsymbol{\Phi}} \hat{\boldsymbol{\theta}}$ by normalized Wendland C^2 CS-RBFs

$$\hat{\Phi}_{\ell q} = \frac{\phi_q(\mathbf{x}_\ell)}{\sum_s \phi_s(\mathbf{x}_\ell)}, \quad \phi_q(\mathbf{x}) = (4r + 1) [\max(0, 1 - r)]^4, \quad r = \frac{\sqrt{\|\mathbf{x} - \mathbf{x}_q\|^2 + \varrho^2}}{r_s}, \quad (10)$$

with support radius r_s and a small ϱ regularizing the spatial derivative at $r \rightarrow 0$. Normalization makes each row of $\hat{\boldsymbol{\Phi}}$ sum to one, so the bound $[-\frac{\pi}{2}, \frac{\pi}{2}]$ is preserved and $\boldsymbol{\theta}$ is C^1 . The un-normalized radial derivative is

$$\phi_{q,x}(\mathbf{x}) = \frac{\partial \phi_q}{\partial x} = -\frac{20(x - x_q)}{r_s^2} [\max(0, 1 - r)]^3, \quad (11)$$

and likewise $\phi_{q,y}$; these follow from $\phi'(r) = -20r(1 - r)^3$ and $\partial r / \partial x = (x - x_q)/(r_s^2 r)$. The normalized derivative operators follow from the quotient rule,

$$\hat{\Phi}_{,x} = \mathbf{S}^{-1} \phi_{,x} - \text{diag}(\mathbf{S}_{,x}) \mathbf{S}^{-2} \phi, \quad \mathbf{S} = \sum_s \phi_s, \quad \mathbf{S}_{,x} = \sum_s \phi_{s,x}, \quad (12)$$

(with ϕ the un-normalized CS-RBF matrix), and analogously $\hat{\Phi}_{,y}$. These sparse $N \times M$ operators give the orientation-field spatial gradient $\nabla \boldsymbol{\theta} = (\hat{\Phi}_{,x} \hat{\boldsymbol{\theta}}, \hat{\Phi}_{,y} \hat{\boldsymbol{\theta}})$ used by the curvature constraint (§9). Support-point discretizations are much coarser than the mesh ($M \ll N$).

6 Wave-projection toolpath generation

For the quasi-static CFRP problem the smooth orientation field is converted, in each iteration, into an explicit fiber/matrix material state $\hat{\chi} \in [0, 1]^N$ (1 = fiber, 0 = matrix) by the wave-projection analogy of Rumpf & Pazos and Ren et al. [3]. A density-dependent spatial weight $\alpha_\ell = H(z_\ell; \eta_\alpha, \beta)$ (smooth Heaviside, below) degrades the phase reconstruction in void regions. The target phase gradient is

$$\nabla \psi_\ell = (-\alpha_\ell \sin \theta_\ell, \alpha_\ell \cos \theta_\ell), \quad (13)$$

i.e. $\nabla \psi \perp$ fibers. With centered finite-difference operators $\mathbf{A}_x, \mathbf{A}_y$ (one-sided at boundaries), the phase is recovered in a least-squares sense $\nabla \psi \approx \mathbf{A} \psi$ via the normal equations

$$\mathbf{G} \psi = \mathbf{A}_x^\top \mathbf{g}_x + \mathbf{A}_y^\top \mathbf{g}_y, \quad \mathbf{G} = \mathbf{A}_x^\top \mathbf{A}_x + \mathbf{A}_y^\top \mathbf{A}_y, \quad (14)$$

where $\mathbf{G}$ is constant and prefactorized once (its single constant null vector removed by pinning one anchor element, $\psi_{\text{anchor}} = 0$). A cosine wave of wavelength d and a second smooth Heaviside produce the towpreg and fiber states

$$\chi_\ell = \frac{1}{2} + \frac{1}{2} \cos\left(\frac{2\pi}{d}(\psi_\ell - \psi_0)\right), \quad \hat{\chi}_\ell = H(\chi_\ell; \eta, \beta), \quad (15)$$

where the smooth Heaviside and its derivative are

$$H(x; \eta, \beta) = \frac{\tanh(\beta\eta) + \tanh(\beta(x - \eta))}{\tanh(\beta\eta) + \tanh(\beta(1 - \eta))}, \quad H'(x) = \frac{\beta(1 - \tanh^2\beta(x - \eta))}{\tanh(\beta\eta) + \tanh(\beta(1 - \eta))}. \quad (16)$$

The wavelength d sets the towpreg width and η the fiber volume fraction within a towpreg, providing process-specific control (AFP vs. MEX).

Discrete adjoint. Given an upstream sensitivity $\partial f / \partial \hat{\chi}$, the chain rule through (15)–(14) yields, with $\partial \chi_\ell / \partial \psi_\ell = -\frac{\pi}{d} \sin(\frac{2\pi}{d}(\psi_\ell - \psi_0))$,

$$\boldsymbol{\lambda}_\psi = \mathbf{G}^{-1} \left(\frac{\partial f}{\partial \hat{\chi}} \odot H'(\boldsymbol{\chi}) \odot \frac{\partial \boldsymbol{\chi}}{\partial \boldsymbol{\psi}} \right), \quad \mathbf{a}_x = \mathbf{A}_x \boldsymbol{\lambda}_\psi, \quad \mathbf{a}_y = \mathbf{A}_y \boldsymbol{\lambda}_\psi, \quad (17)$$

(the pinned solve reuses the prefactorization of $\mathbf{G}$), and finally

$$\frac{\partial f}{\partial z_\ell} = (-a_{x,\ell} \sin \theta_\ell + a_{y,\ell} \cos \theta_\ell) H'(z_\ell), \quad \frac{\partial f}{\partial \theta_\ell} = -\alpha_\ell (a_{x,\ell} \cos \theta_\ell + a_{y,\ell} \sin \theta_\ell). \quad (18)$$

Equations (17)–(18) were verified against finite differences to 4–7 digits.

Part II: Physics, objectives, and sensitivities

7 Quasi-static compliance and its adjoint

The global stiffness is assembled from the two-phase element matrices

$$\mathbf{K}(\mathbf{z}, \boldsymbol{\theta}) = \sum_\ell E_\ell \mathbf{k}_\ell^0, \quad \mathbf{k}_\ell^0 = \hat{\chi}_\ell \mathbf{k}_f(\theta_\ell) + (1 - \hat{\chi}_\ell) \mathbf{k}_m, \quad (19)$$

and the state solves $\mathbf{K}\mathbf{u} = \mathbf{F}$; the objective is compliance $f = \mathbf{F}^\top \mathbf{u}$, which is self-adjoint, $\partial f / \partial s = -\mathbf{u}^\top (\partial \mathbf{K} / \partial s) \mathbf{u}$. Collecting the element energy $e_\ell[\mathbf{k}] = \mathbf{u}_\ell^\top \mathbf{k} \mathbf{u}_\ell$, the full sensitivity chain is

$$\frac{\partial f}{\partial E_\ell} = -e_\ell[\mathbf{k}_\ell^0], \quad \frac{\partial f}{\partial \mathbf{z}} \Big|_{\text{SIMP}} = \mathbf{P}^\top \left[\frac{\partial f}{\partial \mathbf{E}} \odot p(1 - \varepsilon) \mathbf{y}^{p-1} \right], \quad (20)$$

$$\frac{\partial f}{\partial \hat{\chi}_\ell} = -E_\ell e_\ell[\mathbf{k}_f - \mathbf{k}_m], \quad (\text{then wave adjoint (17)–(18)}) \Rightarrow \frac{\partial f}{\partial \mathbf{z}} \Big|_\chi, \quad \frac{\partial f}{\partial \boldsymbol{\theta}} \Big|_\chi, \quad (21)$$

$$\frac{\partial f}{\partial \theta_\ell} \Big|_{\text{dir}} = -E_\ell \hat{\chi}_\ell e_\ell \left[\frac{d\mathbf{k}_f}{d\theta} \right], \quad \frac{\partial f}{\partial \boldsymbol{\theta}} = \hat{\boldsymbol{\Phi}}^\top \left(\frac{\partial f}{\partial \boldsymbol{\theta}} \Big|_{\text{dir}} + \frac{\partial f}{\partial \boldsymbol{\theta}} \Big|_\chi \right), \quad (22)$$

with $\partial f / \partial \mathbf{z} = \partial f / \partial \mathbf{z} \Big|_{\text{SIMP}} + \partial f / \partial \mathbf{z} \Big|_\chi$. The derivative $d\mathbf{k}_f / d\theta$ comes for free from (8).

8 Volume constraints

The composite- and fiber-volume constraints and their gradients are

$$v_g = \frac{\sum_{\ell} A_{\ell} y_{\ell}}{\sum_{\ell} A_{\ell}} - v_{\text{all}}, \quad v_{fg} = \frac{\sum_{\ell} A_{\ell} y_{\ell} \hat{\chi}_{\ell}}{\sum_{\ell} A_{\ell}} - v_{f,\text{all}}, \quad (23)$$

$\partial v_g / \partial \mathbf{z} = \mathbf{P}^{\top}(\mathbf{A} / \sum A)$, and for v_{fg} the density path $\partial v_{fg} / \partial \mathbf{y} = \mathbf{A} \odot \hat{\chi} / \sum A$ plus the material-state path $\partial v_{fg} / \partial \hat{\chi} = \mathbf{A} \odot \mathbf{y} / \sum A$ routed through the wave adjoint to give $\partial v_{fg} / \partial \mathbf{z}$ and $\partial v_{fg} / \partial \hat{\theta}$.

9 Fiber-curvature (curl) constraint

Fiber-path curvature is limited by constraining the out-of-plane curl of the unit orientation field $\mathbf{v} = (\cos \theta, \sin \theta)$:

$$\zeta_{\ell} = \cos \theta_{\ell} (\hat{\Phi}_{,x} \hat{\theta})_{\ell} + \sin \theta_{\ell} (\hat{\Phi}_{,y} \hat{\theta})_{\ell}, \quad \zeta_{\ell}^2 \leq \zeta_{\text{all}}^2 \quad (\ell = 1, \dots, N). \quad (24)$$

Because $\theta = \hat{\Phi} \hat{\theta}$ is C^1 , the Jacobian is analytic:

$$\frac{\partial \zeta_{\ell}}{\partial \hat{\theta}_q} = (-\sin \theta_{\ell} a_{\ell} + \cos \theta_{\ell} b_{\ell}) \hat{\Phi}_{\ell q} + \cos \theta_{\ell} \hat{\Phi}_{,x, \ell q} + \sin \theta_{\ell} \hat{\Phi}_{,y, \ell q}, \quad (25)$$

with $a = \hat{\Phi}_{,x} \hat{\theta}$, $b = \hat{\Phi}_{,y} \hat{\theta}$. The N local constraints are handled by the augmented-Lagrangian scheme of §14; (25) was verified to $\sim 10^{-11}$.

10 Time-harmonic elastodynamics

For wave control the state at angular frequency ω solves the complex-symmetric system

$$\mathbf{D}(\theta) \mathbf{u} = \mathbf{F}, \quad \mathbf{D} = (1 + i\eta) \mathbf{K}(\theta) - \omega^2 \mathbf{M} + i\omega \mathbf{C}_{\text{sponge}}, \quad (26)$$

with structural damping η , consistent mass $\mathbf{M}$, and a mass-proportional absorbing layer $\mathbf{C}_{\text{sponge}} = \sum_{\ell} c_{\ell} \mathbf{m}_{\ell}$. The sponge coefficient ramps quadratically toward chosen open boundaries, $c_{\ell} = c_0 [\max(0, d_{\ell})]^2$ with d_{ℓ} the normalized penetration into a boundary margin, so that outgoing waves are attenuated before reflecting (emulating an unbounded medium). For the pure wave-steering studies the density is fixed solid ($\mathbf{z} \equiv 1$, $\mathbf{K} = \sum_{\ell} \mathbf{k}_f(\theta_{\ell})$), so *only* the fiber orientation is designed.

10.1 Energy-localization objective and adjoint

Let $\mathbf{L}$ be a real diagonal operator selecting a focus region. The focus energy $E = \mathbf{u}^{\text{H}} \mathbf{L} \mathbf{u}$ is real; differentiating $\mathbf{u} = \mathbf{D}^{-1} \mathbf{F}$ and using $d\mathbf{u} = -\mathbf{D}^{-1} (d\mathbf{D}) \mathbf{u}$ gives $dE = -2 \operatorname{Re}(\mathbf{u}^{\text{H}} \mathbf{L} \mathbf{D}^{-1} d\mathbf{D} \mathbf{u})$. Introducing the adjoint field $\boldsymbol{\lambda}$ solving the conjugate system (exploiting $\mathbf{D}^{\top} = \mathbf{D}$, hence $\mathbf{D}^{\text{H}} = \overline{\mathbf{D}}$),

$$\overline{\mathbf{D}} \boldsymbol{\lambda} = \mathbf{L} \mathbf{u}, \quad \frac{\partial E}{\partial \theta_{\ell}} = -2 \operatorname{Re} \left[\boldsymbol{\lambda}_{\ell}^{\text{H}} (1 + i\eta) \frac{d\mathbf{k}_f(\theta_{\ell})}{d\theta} \mathbf{u}_{\ell} \right], \quad (27)$$

since only $\mathbf{K}$ (through $(1 + i\eta)$) depends on θ . The support-point gradient is $\hat{\Phi}^{\top} (\partial E / \partial \theta)$. Equation (27) was verified to $\leq 10^{-7}$.

10.2 Multi-frequency (broadband) max–min objective

For frequencies $\{\omega_i\}_{i=1}^{n_f}$ with energies E_i and straight-fiber baselines E_i^0 , define normalized gains $g_i = E_i/E_i^0$. A robust broadband lens maximizes the worst-case gain through a Kreisselmeier–Steinhauser soft-minimum

$$\max_{\boldsymbol{\theta}} \text{softmin}_i(g_i), \quad \text{softmin}_i(g_i) = -\frac{1}{\rho} \ln \sum_i e^{-\rho g_i}, \quad w_i = \frac{e^{-\rho g_i}}{\sum_j e^{-\rho g_j}}, \quad (28)$$

whose gradient is the soft-min-weighted combination of single-frequency gradients: $\partial_{\boldsymbol{\theta}} \text{softmin} = \sum_i w_i \partial_{\boldsymbol{\theta}} g_i$. Each $\partial_{\boldsymbol{\theta}} E_i$ requires one forward solve (26) and one adjoint solve (27) at ω_i . The alternative **sum** objective replaces w_i by unity. The soft-min form prevents starving one frequency to over-serve another; (28) was verified to $\leq 10^{-7}$ in both modes.

10.3 Elastic cloaking objective

For a plate containing a fixed void (a density hole, $z = 0$ inside), the cloak matches the field to a void-free reference $\mathbf{u}_{\text{ref}}$ in an observation region $\mathbf{W}$:

$$J = (\mathbf{u} - \mathbf{u}_{\text{ref}})^H \mathbf{W} (\mathbf{u} - \mathbf{u}_{\text{ref}}), \quad \bar{\mathbf{D}} \boldsymbol{\lambda} = \mathbf{W} (\mathbf{u} - \mathbf{u}_{\text{ref}}), \quad (29)$$

and $\partial J / \partial \theta_\ell$ has the identical form (27) with $\mathbf{L}\mathbf{u}$ replaced by $\mathbf{W}(\mathbf{u} - \mathbf{u}_{\text{ref}})$; only the fiber orientation in an annular shell around the void is designed. Verified to $\sim 10^{-8}$.

11 Periodic (Bloch) analysis

11.1 Bloch reduction and the generalized eigenproblem

For a wavevector $\mathbf{k}$ the displacement obeys $\mathbf{u}(\mathbf{x} + \mathbf{a}) = e^{i\mathbf{k} \cdot \mathbf{a}} \mathbf{u}(\mathbf{x})$. Assembling $\mathbf{K}, \mathbf{M}$ on the open cell and imposing periodicity by a complex reduction matrix $\mathbf{T}(\mathbf{k})$ (each boundary node slaved to its interior master with phase $e^{i\mathbf{k} \cdot \mathbf{a}}$) yields the Hermitian generalized eigenproblem

$$\mathbf{K}_r(\mathbf{k}) \boldsymbol{\varphi} = \omega^2 \mathbf{M}_r(\mathbf{k}) \boldsymbol{\varphi}, \quad \mathbf{K}_r = \mathbf{T}^H \mathbf{K} \mathbf{T}, \quad \mathbf{M}_r = \mathbf{T}^H \mathbf{M} \mathbf{T}. \quad (30)$$

Lowest bands use a dense Hermitian solver; a few interior bands near a target $\sigma = \omega_\star^2$ use sparse shift-invert (Lanczos), with the eigenvectors re-normalized to $\boldsymbol{\varphi}^H \mathbf{M}_r \boldsymbol{\varphi} = 1$ (giving $\sim 10\times$ speedup for the valley studies). The two-phase cell interpolates $\mathbf{k}_\ell^0 = z_\ell^p \mathbf{k}_f(\theta_\ell) + (1 - z_\ell^p) \mathbf{k}_m$ and $\rho_\ell = z_\ell \rho_f + (1 - z_\ell) \rho_m$.

11.2 Eigenvalue sensitivities (Hellmann–Feynman)

For a simple eigenvalue $\lambda = \omega^2$ with $\mathbf{M}_r$ -normalized $\boldsymbol{\varphi}$ and full-space vector $\boldsymbol{\phi} = \mathbf{T} \boldsymbol{\varphi}$ (with $\mathbf{T}$ design-independent),

$$\frac{\partial \lambda}{\partial z_\ell} = p z_\ell^{p-1} \boldsymbol{\phi}_\ell^H (\mathbf{k}_f - \mathbf{k}_m) \boldsymbol{\phi}_\ell - \lambda (\rho_f - \rho_m) \boldsymbol{\phi}_\ell^H \mathbf{m}_\ell \boldsymbol{\phi}_\ell, \quad \frac{\partial \lambda}{\partial \theta_\ell} = z_\ell^p \boldsymbol{\phi}_\ell^H \frac{d\mathbf{k}_f}{d\theta} \boldsymbol{\phi}_\ell, \quad (31)$$

and $\partial \omega / \partial (\cdot) = \partial \lambda / \partial (\cdot) / (2\omega)$, with the support-point orientation gradient $\hat{\boldsymbol{\Phi}}^T (\partial \lambda / \partial \boldsymbol{\theta})$. Verified to $\sim 10^{-6}$ across bands.

11.3 Berry curvature: Fukui–Hatsugai–Suzuki

For an isolated band we compute the (gauge-invariant) discrete Berry curvature on an $n_k \times n_k$ grid over the Brillouin zone. Using the $\mathbf{M}$ -weighted inner product of full-space Bloch states $\phi(\mathbf{k}) = \mathbf{T}(\mathbf{k})\varphi(\mathbf{k})$, form directed link variables and plaquette field strength

$$U_\mu(\mathbf{k}) = \langle \phi(\mathbf{k}) | \mathbf{M} | \phi(\mathbf{k} + \mathbf{e}_\mu) \rangle, \quad F(\mathbf{k}) = \text{Im} \ln [U_x(\mathbf{k}) U_y(\mathbf{k} + \mathbf{e}_x) U_x(\mathbf{k} + \mathbf{e}_y)^* U_y(\mathbf{k})^*]. \quad (32)$$

Any $\mathbf{k}$ -local phase of ϕ cancels around the loop, so F is gauge invariant. The Chern number is $C = \frac{1}{2\pi} \sum_{\mathbf{k}} F$ and the *valley* Chern is the half-Brillouin-zone sum about K or K' . The estimator was validated on the Qi–Wu–Zhang model, recovering the exact $C \in \{0, \pm 1\}$ across its phase transitions.

11.4 Band-gap objectives

Directional and complete gaps between bands $p, p+1$ over a $\mathbf{k}$ -sample $\mathcal{S}$ use smooth KS aggregation

$$\text{gap} = \text{softmin}_{\mathbf{k} \in \mathcal{S}} \omega_{p+1}(\mathbf{k}) - \text{softmax}_{\mathbf{k} \in \mathcal{S}} \omega_p(\mathbf{k}), \quad \text{softmax}_\rho(v) = \frac{1}{\rho} \ln \sum e^{\rho v}, \quad (33)$$

with gradients the KS-weighted eigenvalue sensitivities (36). The directional band-gap optimizer adds an orthogonal-direction penalty $\text{relu}(\text{gap}_\perp)$ to keep the gap direction-selective; the valley-gap optimizer adds an isolation penalty $\text{relu}(m - (\omega_p - \omega_{p-1})) + \text{relu}(m - (\omega_{p+2} - \omega_{p+1}))$ to keep the Dirac pair isolated; both add a volume penalty $c(\text{vol} - \text{vol}_\star)^2$.

Part III: Optimization algorithms

12 Method of Moving Asymptotes (bound-constrained)

All problems here fold their constraints into the objective (augmented Lagrangian or explicit penalties), so we use MMA with $m = 0$ explicit constraints, whose separable convex subproblem has a closed-form primal solution. At design $\mathbf{x}$ with objective gradient $\partial_x f_0$, MMA maintains moving asymptotes $\mathbf{L}, \mathbf{U}$ that bracket $\mathbf{x}$. On iteration t :

Algorithm 1 MMA update (MMA.update)

- 1: **asymptotes:** if $t \leq 2$: $\mathbf{L} = \mathbf{x} - s_0 \text{span}$, $\mathbf{U} = \mathbf{x} + s_0 \text{span}$; else $\gamma = \{1.2, 0.7, 1\}$ by $\text{sign}[(\mathbf{x} - \mathbf{x}^{-1}) \odot (\mathbf{x}^{-1} - \mathbf{x}^{-2})]$,
 $\mathbf{L} = \mathbf{x} - \gamma \odot (\mathbf{x}^{-1} - \mathbf{L}^{-1})$, $\mathbf{U} = \mathbf{x} + \gamma \odot (\mathbf{U}^{-1} - \mathbf{x}^{-1})$, then clip to $[\mathbf{x} - 10 \text{span}, \mathbf{x} + 0.01 \text{span}]$
etc.
 - 2: **move limits:** $\alpha = \max(\mathbf{x}_{\min}, \mathbf{L} + 0.1(\mathbf{x} - \mathbf{L}), \mathbf{x} - \text{mv span})$, $\beta = \min(\mathbf{x}_{\max}, \mathbf{U} - 0.1(\mathbf{U} - \mathbf{x}), \mathbf{x} + \text{mv span})$
 - 3: **convex coefficients:** with $\partial_x f^\pm = \max(\pm \partial_x f_0, 0)$ and $r = r_0 / \text{span}$,
 $\mathbf{p}_0 = (\mathbf{U} - \mathbf{x})^2 (1.001 \partial_x f^+ + 0.001 \partial_x f^- + r)$, $\mathbf{q}_0 = (\mathbf{x} - \mathbf{L})^2 (0.001 \partial_x f^+ + 1.001 \partial_x f^- + r)$
 - 4: **closed-form minimizer:** $\mathbf{x}^{\text{new}} = \frac{\sqrt{\mathbf{p}_0} \odot \mathbf{L} + \sqrt{\mathbf{q}_0} \odot \mathbf{U}}{\sqrt{\mathbf{p}_0} + \sqrt{\mathbf{q}_0}}$, then clip to $[\alpha, \beta]$
-

The step-4 formula is the exact stationary point of the separable MMA approximation $\sum_j [p_{0j}/(U_j - x_j) + q_{0j}/(x_j - L_j)]$, obtained by setting each coordinate derivative to zero. The small mixing (0.001) and r_0 regularization keep $\mathbf{p}_0, \mathbf{q}_0 > 0$.

13 Augmented Lagrangian for local constraints

For the many local curl constraints, the outer loop solves a sequence of bound-constrained subproblems (each MMAIter MMA steps) with the augmented-Lagrangian objective

$$\mathcal{L} = f + \underbrace{\sum_{j \in \text{global}} \left[\lambda_j h_j + \frac{\mu}{2} h_j^2 \right] + \frac{1}{N} \sum_{j \in \text{local}} \left[\lambda_j h_j + \frac{\mu}{2} h_j^2 \right]}_J, \quad h_j = \max\left(g_j, -\frac{\lambda_j}{\mu}\right), \quad (34)$$

where the $1/N$ scaling makes the local-constraint penalty independent of mesh size. Its gradient is $\partial_{\mathbf{W}} \mathcal{L} = \partial_{\mathbf{W}} f + \sum_j (\lambda_j + \mu h_j) \partial_{\mathbf{W}} g_j \mathbf{1}[g_j > -\lambda_j/\mu]$, with $\partial_{\mathbf{W}} g_j$ the volume/curl gradients of §8–9. After each subproblem the multipliers and penalty update as $\lambda_j \leftarrow \max(\lambda_j + \mu h_j, 0)$ and $\mu \leftarrow \min(\xi \mu, \mu_{\max})$.

Algorithm 2 Augmented-Lagrangian driver (optimize_cfrp)

```

1: init  $\mathbf{z}, \hat{\boldsymbol{\theta}}; \boldsymbol{\lambda} = \boldsymbol{\lambda}_0; \mu = \mu_0; \text{best} = \infty$ 
2: for  $k = 0, \dots, \text{MaxOuter}$  do
3:   for MMAIter steps do ▷ inner: minimize  $\mathcal{L}$ 
4:     toolpath generation (wave projection (15)); solve  $\mathbf{K}\mathbf{u} = \mathbf{F}$ ; evaluate  $f, \mathbf{g}, \boldsymbol{\zeta}$ 
5:      $\mathbf{W} \leftarrow \text{MMA.UPDATE}(\mathbf{W}, \partial_{\mathbf{W}} \mathcal{L})$ 
6:   end for
7:   evaluate feasibility (volume,  $\max |\boldsymbol{\zeta}|/\zeta_{\text{all}}$ ); if feasible and  $f < \text{best}$ , store best
8:    $\lambda_j \leftarrow \max(\lambda_j + \mu h_j, 0)$ ; if infeasible then  $\mu \leftarrow \min(\xi \mu, \mu_{\max})$  ▷ gate penalty growth
9:   if  $\Delta \mathbf{W} < \text{tolD}$  and curl feasible then break; if stale then break
10: end for
11: return best feasible design

```

Two safeguards are essential in practice and are highlighted in Algorithm 2. First, the penalty μ is grown *only while a constraint is violated*; continuing to inflate μ after feasibility ill-conditions the MMA subproblem and destabilizes an already-converged design. Second, the driver returns the best *feasible* iterate rather than the last, since MMA on the coupled density–orientation problem oscillates at the few-percent level rather than reaching a tight design-change tolerance.

Part IV: Results

14 Numerical results

14.1 Validation: curvature-constrained CFRP cantilever

We reproduce the no-void cantilever of Wong et al. [1] at their 240×150 resolution (Table 1). Without a curl constraint the optimizer recovers a two-strut design; imposing $\zeta_{\text{all}} = 2 \text{ m}^{-1}$ drives the maximum fiber curvature to the allowable limit at a compliance cost, matching the reported behavior. Figure 1 shows the clean (filtered/projected) topologies, fiber streamlines, explicit towpregs, and curl maps.

Table 1: No-void cantilever: present results vs. Wong et al. [1].

	f (ours)	f (paper)	$ \zeta _{\max}$ ours	$ \zeta _{\max}$ paper
no curl constraint	0.452	0.42	10.7	8.04
$\text{curl} \leq 2 \text{ m}^{-1}$	0.577	0.49	1.93	2.00

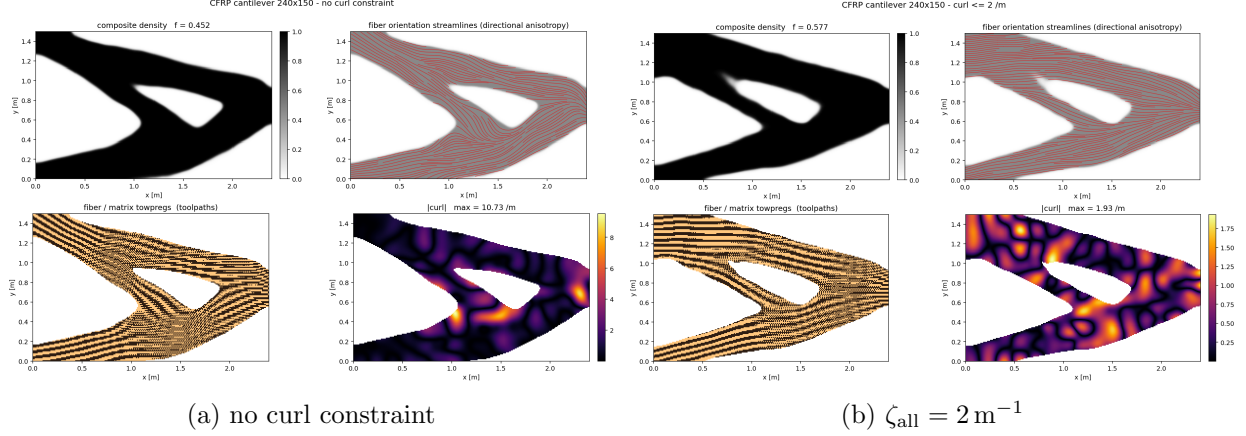


Figure 1: Curvature-constrained CFRP cantilever (density, fiber streamlines, fiber/matrix toolpaths, and $|\zeta|$ map). The curl constraint removes the high-curvature hot spots and enlarges the load-path turning radii.

14.2 Wave-energy localization (single frequency)

A solid composite plate is driven by a plane pressure wave from the left edge; only the fiber orientation is optimized to maximize elastic energy in a small box on the right. The optimized fibers curve into a converging lens and concentrate the field at the target with a **20.6** $\times$ energy gain over the straight-fiber baseline (Figure 2). The material is homogeneous everywhere — only the fiber *direction* varies.

14.3 Broadband localization (multi-frequency)

A lens optimized for a single frequency is narrow-band: optimizing only the centre frequency $\omega = 38$ yields a $49\times$ gain there but *degrades* the low frequency below the straight-fiber baseline ($0.55\times$ at $\omega = 28$). Optimizing the worst-case gain (28) over $\{28, 38, 48\}$ instead produces a single fiber design that focuses *every* frequency almost equally, with gains $\{13.8, 13.9, 13.8\}\times$ (Figure 3). The max-min objective trades peak single-frequency performance for uniform broadband performance.

14.4 Elastic cloaking of a void

A fixed circular void scatters an incident plane wave, casting a downstream shadow and upstream reflections. Optimizing the fiber orientation in an annular shell to match the void-free reference reduces the observation-region scattering energy by **15.3** $\times$ (Figure 4): the shadow is largely refilled and the reflections suppressed, with the toolpaths visibly wrapping the void. As expected for orientation-only control of a hard scatterer, a residual disturbance remains adjacent to the void, but the far field is substantially restored.

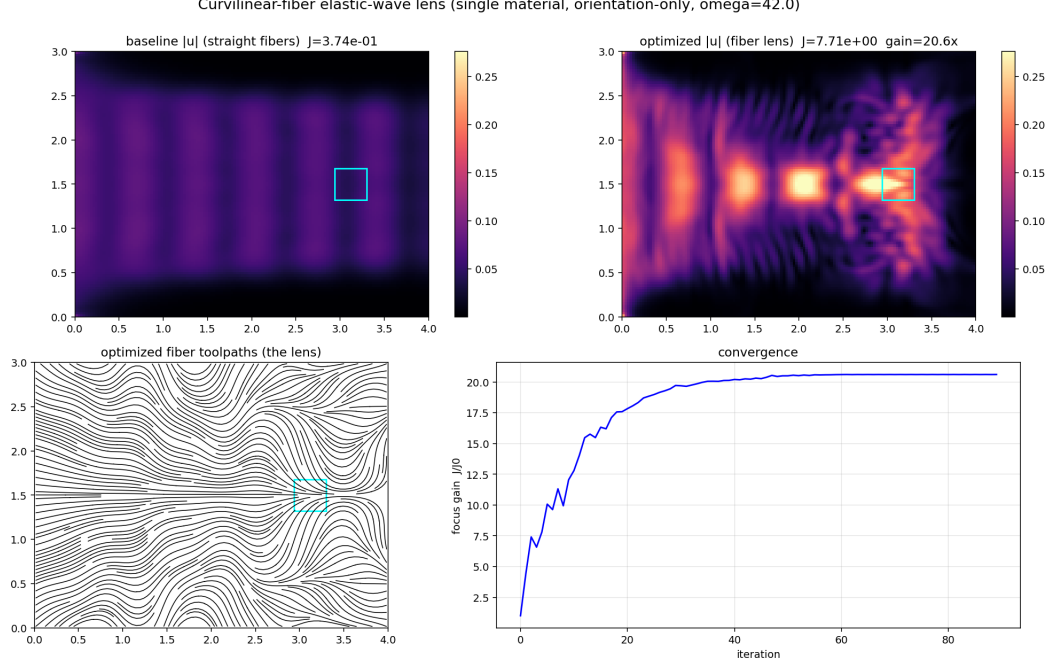


Figure 2: Curvilinear-fiber wave lens (single frequency, orientation-only): baseline vs. optimized field magnitude, the optimized fiber toolpaths (the lens), and convergence of the focus gain to $20.6\times$.

14.5 Topological valley-Hall waveguide (orientation as the Dirac mass)

The same orientation-as-design principle extends to *topological* wave guiding on periodic cells. We use the scalar (antiplane-shear) problem on a triangular lattice (lattice vectors $\mathbf{a}_1 = a(1, 0)$, $\mathbf{a}_2 = a(\frac{1}{2}, \frac{\sqrt{3}}{2})$; inequivalent BZ corners $K = (2\mathbf{b}_1 + \mathbf{b}_2)/3$, $K' = (\mathbf{b}_1 + 2\mathbf{b}_2)/3$ with a single stiff, C_{3v} equilateral-triangle inclusion per cell whose rotation angle α is the design variable.

$k \cdot p$ theory. When the triangle's mirror planes align with the lattice's ($\alpha \equiv 30^\circ$ in our vertex convention) the crystal is C_{3v} at K and the second and third bands meet in a symmetry-protected, *isolated* Dirac cone. Near K the two-band effective Hamiltonian is

$$H(\delta\mathbf{k}) = v_D(\delta k_x \sigma_x + \delta k_y \sigma_y) + m \sigma_z, \quad m(\alpha) \propto \cos(3\alpha), \quad (35)$$

so rotating the triangle away from $\alpha = 30^\circ$ breaks the mirror ($C_{3v} \rightarrow C_3$) and opens the valley gap $2|m|$, with m changing sign across $\alpha = 30^\circ$. The gapped Dirac cone carries Berry curvature $\Omega(\delta\mathbf{k}) = \mp v_D^2 m / 2(v_D^2 |\delta\mathbf{k}|^2 + m^2)^{3/2}$ concentrated at the valley, giving the quantized *valley Chern number* $C_v^K = \frac{1}{2} \text{sgn}(m)$ and $C_v^{K'} = -\frac{1}{2} \text{sgn}(m)$ (total Chern zero, as required by time-reversal symmetry). Two domains with opposite mass (mirror-partner angles α and $60^\circ - \alpha$, e.g. 15° and 45°) differ by $|\Delta C_v| = 1$, so by bulk-edge correspondence their interface hosts one gap-traversing, valley-momentum-locked kink mode per valley.

Design and phase diagram. Cells are meshed with periodic, geometry-conforming, unstructured (gmsh) triangulations — essential, because a structured/sheared mesh systematically breaks C_3 and splits the Dirac cone at a rate that does *not* vanish under refinement. Bands are computed with the Bloch reduction (35). The rod is sized ($R = 0.44a$) and stiffened ($\mu_f/\mu_m = 100$) so that

Broadband toolpath wave lens: one fiber design focuses a band of frequencies

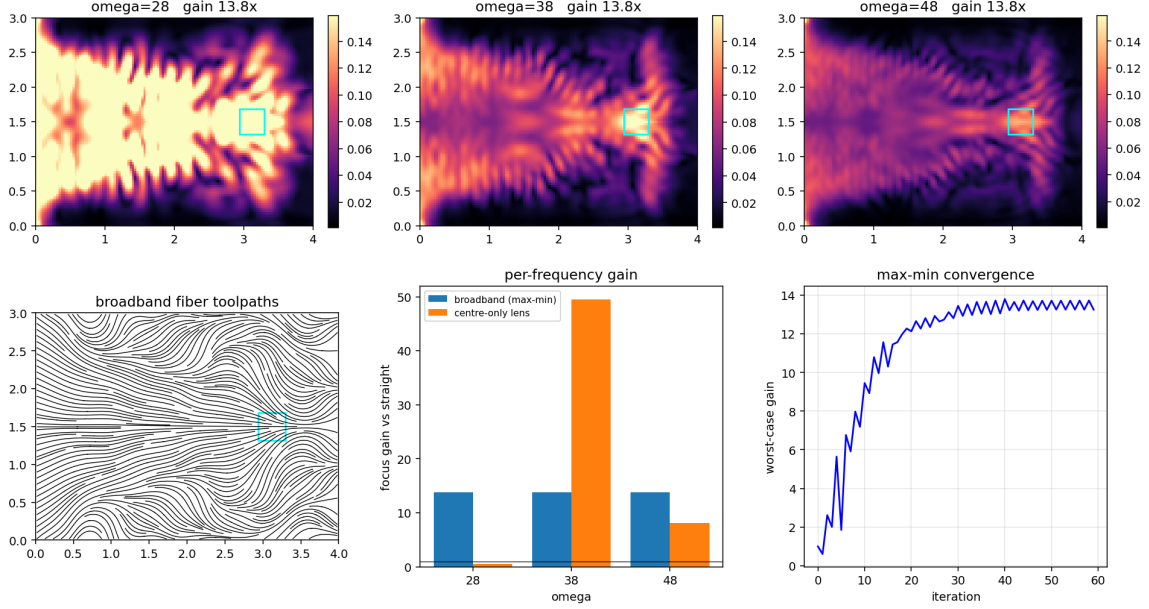


Figure 3: Broadband toolpath lens. Top: optimized field at three frequencies, each focused at the target. Bottom: toolpaths; per-frequency gain of the broadband (max-min) design vs. a narrow-band centre-only lens (which drops below unity at $\omega = 28$); and the worst-case-gain convergence.

the valley pair is isolated (a full gap of 0.60 opens on either side of the Dirac point) with mid-gap frequency $\omega \approx 7.3$. Figure 10 is the resulting topological phase diagram: the complete valley gap closes exactly at the Dirac point $\alpha = 30^\circ$ (the phase boundary) and is maximized away from it — rod orientation is the tunable, optimizable topological mass.

Gradient optimization of the gap, and an honest accounting. The gap is not merely swept but *optimized*. Treating the element density $\mathbf{z}$ (geometry) and the fiber-orientation field $\boldsymbol{\theta}$ (anisotropy) as filtered design fields on the cell, we maximize the complete valley gap

$$\max_{\mathbf{z}, \boldsymbol{\theta}} \left[\text{softmin}_{\mathbf{k} \in \mathcal{S}} \omega_{p+1}(\mathbf{k}) - \text{softmax}_{\mathbf{k} \in \mathcal{S}} \omega_p(\mathbf{k}) \right] - c_v(\text{vol} - \text{vol}_*)^2, \quad (36)$$

over a Brillouin-zone sample $\mathcal{S}$, at fixed volume, by MMA driven by the Hellmann–Feynman sensitivities $\partial\omega/\partial\mathbf{z}$ and $\partial\omega/\partial\boldsymbol{\theta}$ of (36) (extended to the P_1 triangular mesh and finite-difference verified to $\sim 10^{-5}$). Running (41) with the orientation field alone, the density field alone, and both, isolates the mechanisms (Figure 9). The result is a genuine and honest accounting: starting from a seed cell with no complete gap (-0.54), optimizing the *orientation* (anisotropy) alone reaches only -0.08 and optimizing the *density* (geometry) alone reaches only -0.03 — *neither field alone opens a complete gap at fixed volume* — whereas co-optimizing both opens it ($+0.28$). In this anisotropic-fiber medium the two levers are complementary: the fiber orientation is a genuine, necessary ingredient of the optimized topological gap, not a passive accessory. (For an *isotropic* scatterer, geometry alone suffices, as the phase diagram above shows; anisotropy becomes essential precisely when the medium is the fiber composite of interest here.)

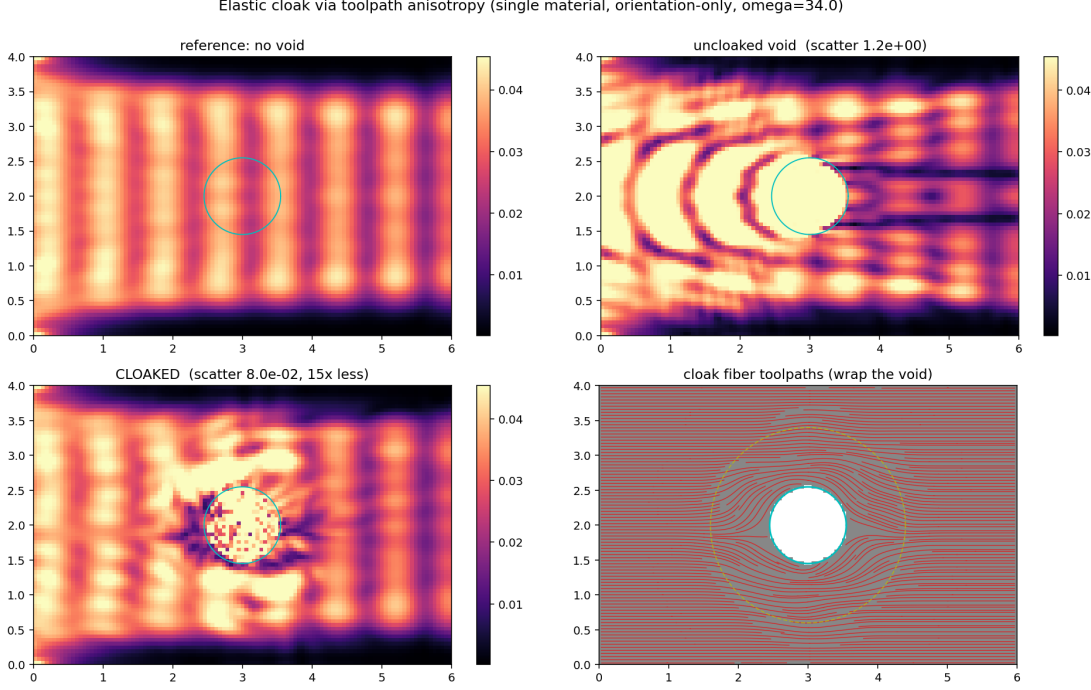


Figure 4: Elastic cloak via toolpath anisotropy: reference (no void), uncloaked void (strong shadow/reflection), cloaked design ($15.3\times$ less scatter), and the fiber toolpaths that wrap the void.

Kink modes and guiding. A ribbon supercell with an $\alpha = 15^\circ \mid 45^\circ$ mirror-partner wall exhibits a single, strongly wall-localized branch that traverses the entire bulk gap, connecting the lower and upper bulk bands through the projected valleys $\kappa = k \cdot \mathbf{a}_2 = \pm 2\pi/3$ (Figure 11) — the direct realization of the bulk–edge correspondence. In a finite domain (Figure 12), a time-harmonic source on a domain wall aligned with the lattice zigzag ($\mathbf{a}_2$) launches this kink mode: at mid-gap the field is tightly confined to the wall and transported to the far end, whereas at a bulk-band frequency it disperses. The wall carries a clean, broadband guided band across the *entire* bulk gap (transmitted energy $\sim 10^1\text{--}10^2$), a factor ~ 1500 above the same source at an in-band frequency and $\sim 10^6$ above the deep-gap (fully reflected) level — the signature of a protected edge channel rather than a resonant defect line.

Scope. Two honest limitations frame this result. First, a passive elastic medium is time-reversal symmetric, so only valley-Hall (not chiral quantum-Hall) transport is available without active components. Second, while straight-wall guiding is robust and broadband, we found that transmission around a *sharp* 120° bend is strongly reduced in this scalar model (the valley polarization, though sufficient to bind and guide the mode, is not strong enough to fully suppress inter-valley backscattering at an abrupt corner); high-fidelity cornering would require further tuning of the rod for maximal valley polarization. The essential point stands: an *orientation* degree of freedom — the same design lever used for the finite-domain devices, and directly analogous to a fiber toolpath angle — realizes a tunable topological mass and a working topological edge waveguide.

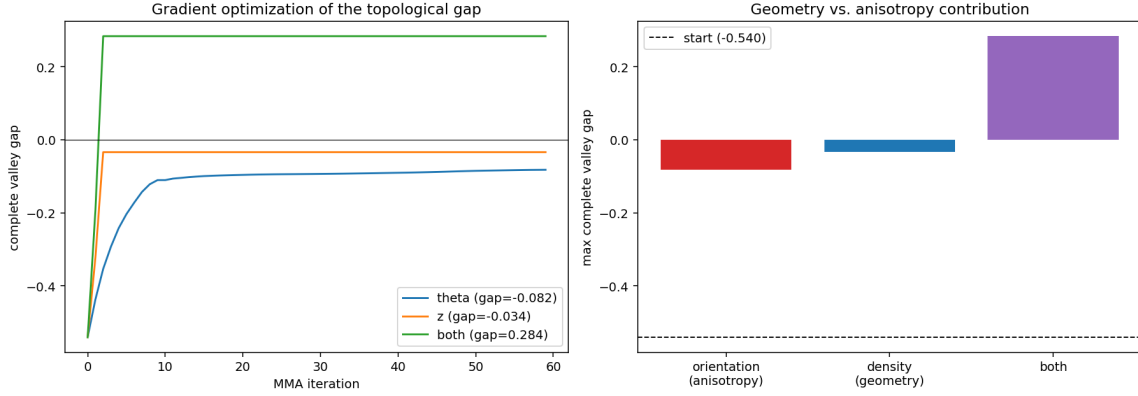


Figure 5: Gradient optimization of the topological valley gap (41) by MMA + Hellmann–Feynman sensitivities. Left: best-feasible complete gap vs. iteration. Right: at fixed volume, orientation-only (anisotropy) and density-only (geometry) each fail to open a complete gap; only co-optimizing both succeeds — an honest measure of the anisotropy’s necessary contribution.

15 Verification

Every gradient in this framework is checked against central finite differences, $\partial_k f \approx [f(\mathbf{x} + \epsilon \mathbf{e}_k) - f(\mathbf{x} - \epsilon \mathbf{e}_k)] / (2\epsilon)$, $\epsilon = 10^{-6}$. Table 2 collects the worst-case relative errors, all at or below the finite-difference truncation floor.

Table 2: Adjoint/Hellmann–Feynman gradients vs. central finite differences.

Routine	Quantity	max. rel. error
Fourier stiffness (8)	$d\mathbf{k}_f / d\theta$	8×10^{-11}
CS-RBF derivatives (12)	$\hat{\Phi}_{,x}, \hat{\Phi}_{,y}$	5×10^{-10}
Wave projection (18)	$\partial \hat{\chi} / \partial(z, \theta)$	$\sim 10^{-6}$
Compliance (§7)	$\partial f / \partial(\mathbf{z}, \hat{\theta})$	$\sim 10^{-6}$
Curl Jacobian (25)	$\partial \zeta / \partial \hat{\theta}$	1×10^{-11}
Focus energy (27)	$\partial E / \partial \hat{\theta}$	6×10^{-7}
Multi-frequency (28)	$\partial_{\hat{\theta}}$ (sum / minmax)	$6 \times 10^{-8} / 2 \times 10^{-7}$
Cloak (§11.3)	$\partial J / \partial \hat{\theta}$	4×10^{-8}
Eigenvalue (36)	$\partial \omega / \partial(\mathbf{z}, \hat{\theta})$	$\sim 10^{-6}$
Berry curvature (37)	Chern (QWZ model)	exact $\{0, \pm 1\}$

16 Conclusion

We have given a complete mathematical and algorithmic description of a unified density-plus-orientation optimization framework for continuous-fiber composites, spanning quasi-static curvature-constrained design and time-harmonic wave control. The exact finite-Fourier representation of the rotated fiber stiffness, the C^1 CS-RBF orientation map with analytic curl, the wave-projection tool-path operator with its discrete adjoint, and the complex-symmetric harmonic adjoint together make

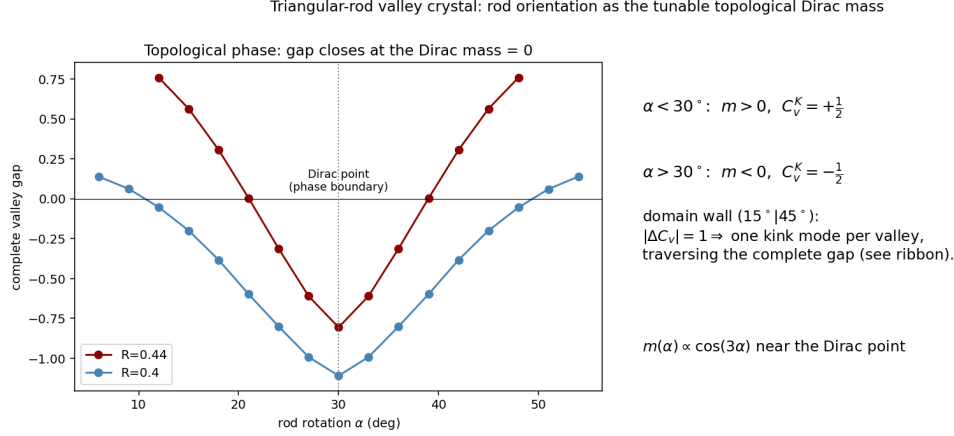


Figure 6: Topological phase diagram of the triangular-rod valley crystal: the complete valley gap (left) closes at the C_{3v} Dirac point $\alpha = 30^\circ$ (mass $m = 0$, phase boundary) and reopens with opposite valley Chern on either side. Rod orientation α is the tunable, optimizable topological Dirac mass; R is the rod size.

orientation an inexpensive, fully differentiable design field. In the finite-domain wave setting, curvilinear fiber toolpaths alone — with a homogeneous base material — focus elastic energy ($> 20\times$), do so across a band of frequencies through a max-min objective ($\sim 14\times$ uniform), and cloak a void ($\sim 15\times$ scatter reduction), all driven by adjoint gradients verified to $\sim 10^{-7}$. In the periodic setting, the orientation of a C_3 -symmetric scatterer acts as a tunable topological Dirac mass whose sign sets the valley Chern number; a mirror-partner domain wall then guides a gap-traversing valley-Hall kink mode, realizing a working topological edge waveguide (mid-gap/in-band transmission ratio $\sim 10^3$) with the same orientation lever used for the finite-domain devices. Robust guiding requires an isolated Dirac cone and a complete gap — achieved here with a C_3 -conforming unstructured mesh and a near-rigid triangular scatterer — and, for sharp 120° corners, strong valley polarization that remains a design target. The manufacturability machinery carries over unchanged, making these designs candidates for additive fabrication.

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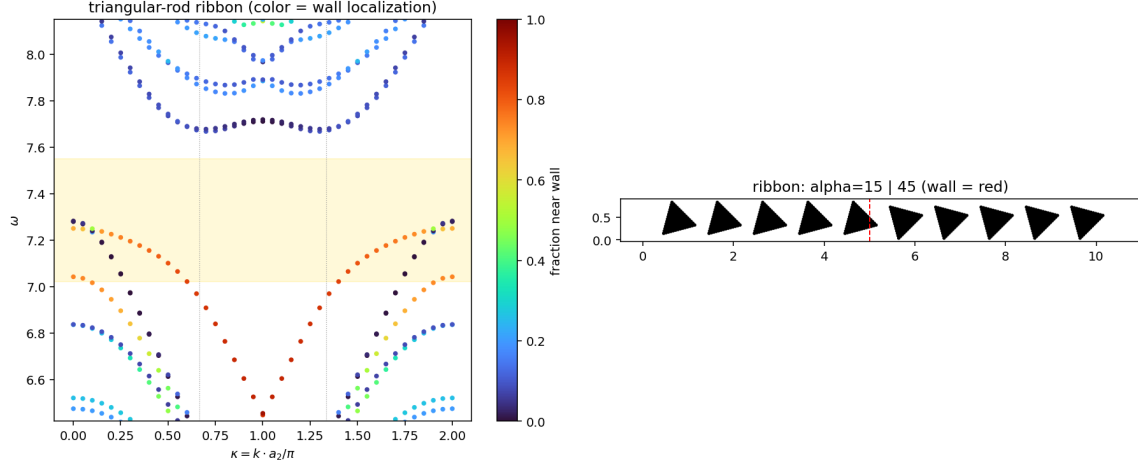


Figure 7: Ribbon band structure (left) of an $\alpha = 15^\circ | 45^\circ$ mirror-partner domain wall (right). A strongly wall-localized (colour = fraction of modal energy at the wall) branch *traverses* the complete bulk gap (shaded) through the projected valleys $\kappa = 2\pi/3, 4\pi/3$: the topological kink mode.

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Triangular-rod valley waveguide around a sharp bend: mid-gap guiding vs in-band (1466x)

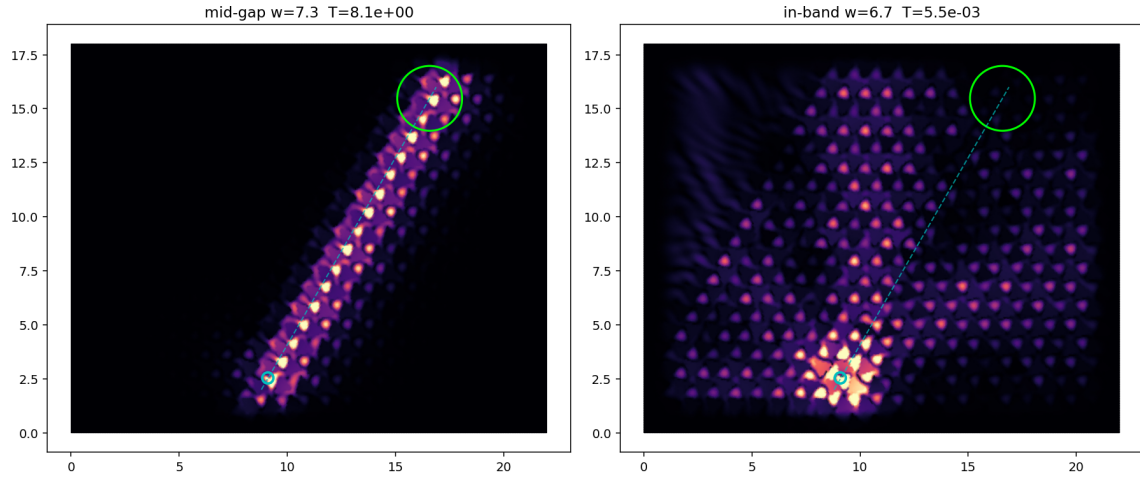


Figure 8: Finite-domain valley waveguide. At mid-gap (left) the kink mode is tightly confined to the lattice-aligned domain wall and transported from source (cyan) to the output (green); at a bulk-band frequency (right) energy disperses into the bulk. The mid-gap/in-band transmission ratio is $\sim 1.5 \times 10^3$.